

position is given by

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_1 + L_2 + L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The screw axes $\mathcal{B}_i = (\omega_i, v_i)$ are listed in the following table:

i	ω_i	v_i
1	$(0, 0, 1)$	$(0, 0, 0)$
2	$(0, 1, 0)$	$(L_1 + L_2 + L_3, 0, 0)$
3	$(0, 0, 1)$	$(0, 0, 0)$
4	$(0, 1, 0)$	$(L_2 + L_3, 0, W_1)$
5	$(0, 0, 1)$	$(0, 0, 0)$
6	$(0, 1, 0)$	$(L_3, 0, 0)$
7	$(0, 0, 1)$	$(0, 0, 0)$

Figure 4.9 shows the WAM arm with $\theta_2 = 45^\circ$, $\theta_4 = -45^\circ$, $\theta_6 = -90^\circ$ and all other joint angles equal to zero, giving

$$T(\theta) = M e^{[\mathcal{B}_2]\pi/4} e^{-[\mathcal{B}_4]\pi/4} e^{-[\mathcal{B}_6]\pi/2} = \begin{bmatrix} 0 & 0 & -1 & 0.3157 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0.6571 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4.2 The Universal Robot Description Format

The **Universal Robot Description Format** (URDF) is an XML (eXtensible Markup Language) file format used by the Robot Operating System (ROS) to describe the kinematics, inertial properties, and link geometry of robots. A URDF file describes the joints and links of a robot:

- **Joints.** Joints connect two links: a **parent** link and a **child** link. A few of the possible joint types include prismatic, revolute (including joint limits), continuous (revolute without joint limits), and fixed (a virtual joint that does not permit any motion). Each joint has an **origin** frame that defines the position and orientation of the **child** link frame relative to the **parent** link frame when the joint variable is zero. The **origin** is on the joint's axis. Each joint has an **axis** 3-vector, a unit vector expressed in the **child** link's frame, in the direction of positive rotation for a revolute joint or positive translation for a prismatic joint.

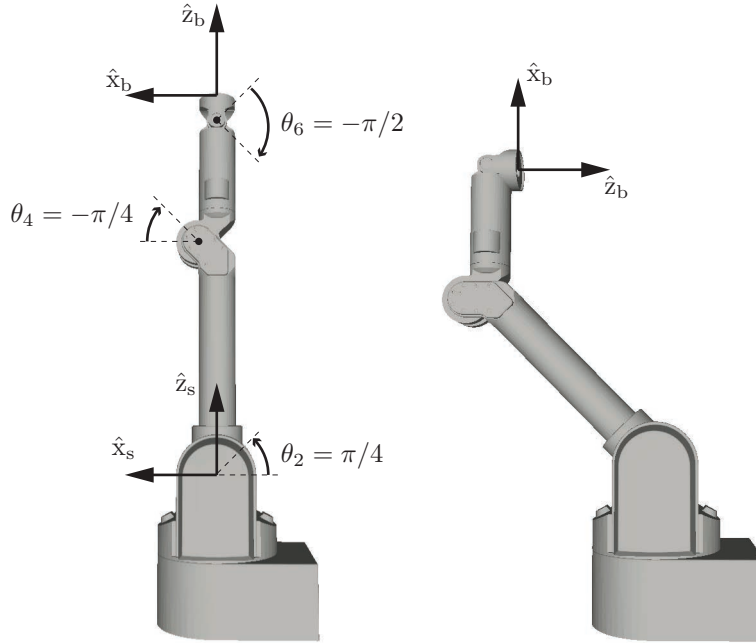


Figure 4.9: (Left) The WAM at its home configuration, with the axes of joints 2, 4, and 6 indicated. (Right) The WAM at $\theta = (\theta_1, \dots, \theta_7) = (0, \pi/4, 0, -\pi/4, 0, -\pi/2, 0)$.

- **Links.** While the joints fully describe the kinematics of a robot, the links define its mass properties. These start to be needed in Chapter 8, when we begin to study the dynamics of robots. The elements of a link include its **mass**; an **origin** frame that defines the position and orientation of a frame at the link's center of mass relative to the link's joint frame described above; and an **inertia** matrix, relative to the link's center of mass frame, specified by the six elements on or above the diagonal. (As we will see in Chapter 8, the inertia matrix for a rigid body is a 3×3 symmetric positive-definite matrix. Since the inertia matrix is symmetric, it is only necessary to define the terms on and above the diagonal.)

Note that most links have two frames rigidly attached: a first frame at the joint (defined by the joint element that connects the link to its parent) and a second frame at the link's center of mass (defined by the link element).

A URDF file can represent any robot with a tree structure. This includes serial-chain robot arms and robot hands, but not a Stewart platform or other

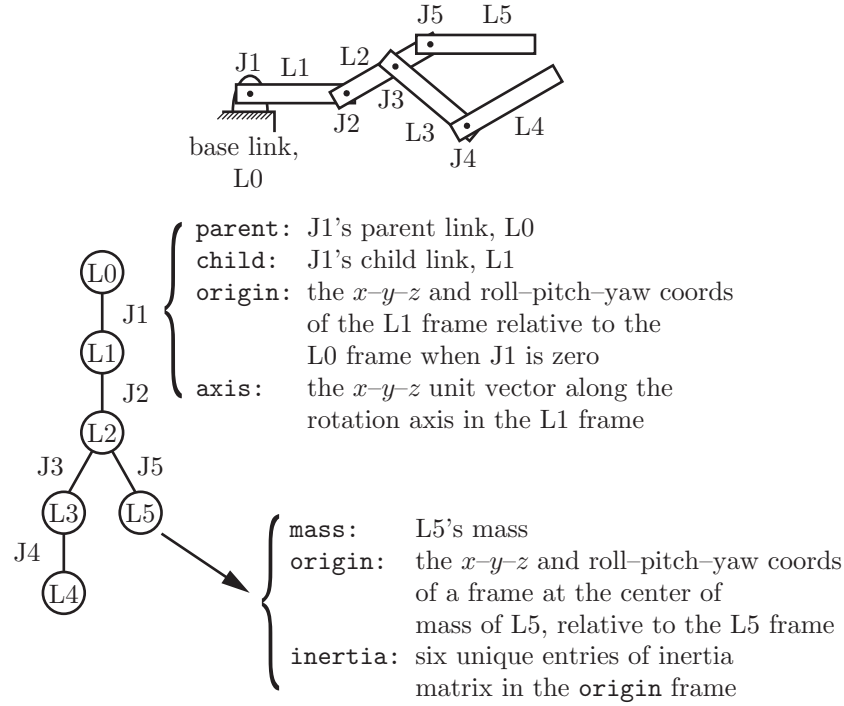


Figure 4.10: A five-link robot represented as a tree, where the nodes of the tree are the links and the edges of the tree are the joints.

mechanisms with closed loops. An example of a robot with a tree structure is shown in Figure 4.10.

The orientation of a frame $\{b\}$ relative to a frame $\{a\}$ is represented using roll-pitch-yaw coordinates: first, a roll about the fixed $\hat{x}_a$ -axis; then a pitch about the fixed $\hat{y}_a$ -axis; then a yaw about the fixed $\hat{z}_a$ -axis.

The kinematics and mass properties of the UR5 robot arm (Figure 4.11) are defined in the URDF file below, which demonstrates the syntax of the joint's elements (**parent**, **child**, **origin**, and **axis**) and the link's elements (**mass**, **origin**, and **inertia**). A URDF requires a frame defined at every joint, so we define frames $\{1\}$ to $\{6\}$ in addition to the fixed base frame $\{0\}$ (i.e., $\{s\}$) and the end-effector frame $\{7\}$ (i.e., $\{b\}$). Figure 4.11 gives the extra information needed to fully write the URDF.

Although the joint types in the URDF are defined as “continuous,” the UR5 joints do in fact have joint limits; they are omitted here for simplicity. The mass

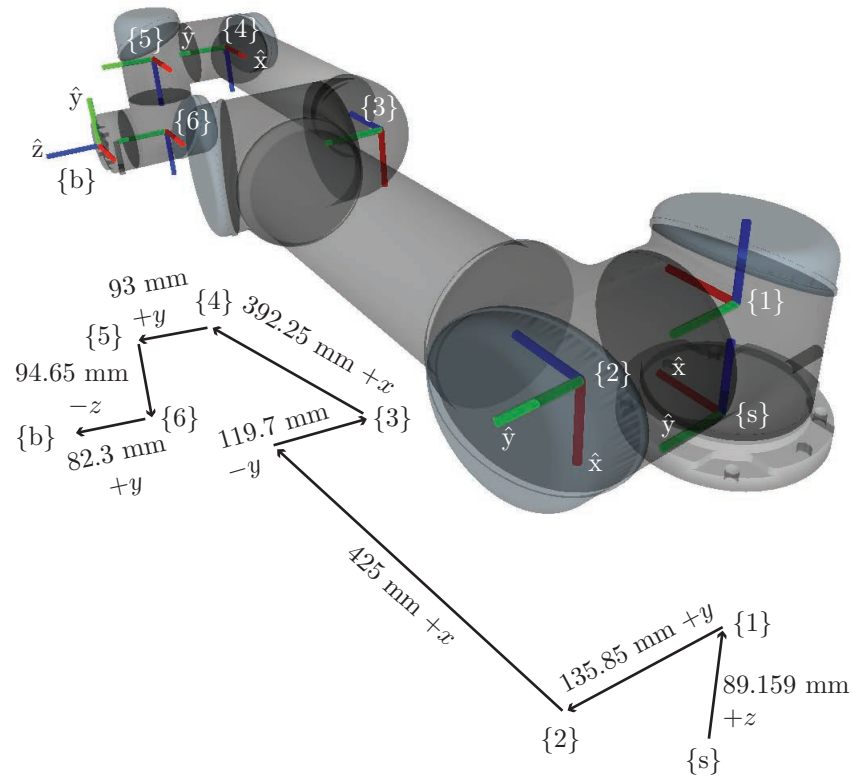


Figure 4.11: The orientations of the frames $\{s\}$ (also called $\{0\}$), $\{b\}$ (also called $\{7\}$), and $\{1\}$ through $\{6\}$ are illustrated on the translucent UR5. The frames $\{s\}$ and $\{1\}$ are aligned with each other; frames $\{2\}$ and $\{3\}$ are aligned with each other; and frames $\{4\}$, $\{5\}$, and $\{6\}$ are aligned with each other. Therefore, only the axes of frames $\{s\}$, $\{2\}$, $\{4\}$, and $\{b\}$ are labeled. Just below the image of the robot is a skeleton indicating how the frames are offset from each other, including distances and directions (expressed in the $\{s\}$ frame).

and inertial properties listed here are not exact.

The UR5 URDF file (kinematics and inertial properties only).

```
<?xml version="1.0" ?>
<robot name="ur5">

<!-- ***** KINEMATIC PROPERTIES (JOINTS) ***** -->
  <joint name="world_joint" type="fixed">
```

```

    <parent link="world"/>
    <child link="base_link"/>
    <origin rpy="0.0 0.0 0.0" xyz="0.0 0.0 0.0"/>
  </joint>
  <joint name="joint1" type="continuous">
    <parent link="base_link"/>
    <child link="link1"/>
    <origin rpy="0.0 0.0 0.0" xyz="0.0 0.0 0.089159"/>
    <axis xyz="0 0 1"/>
  </joint>
  <joint name="joint2" type="continuous">
    <parent link="link1"/>
    <child link="link2"/>
    <origin rpy="0.0 1.570796325 0.0" xyz="0.0 0.13585 0.0"/>
    <axis xyz="0 1 0"/>
  </joint>
  <joint name="joint3" type="continuous">
    <parent link="link2"/>
    <child link="link3"/>
    <origin rpy="0.0 0.0 0.0" xyz="0.0 -0.1197 0.425"/>
    <axis xyz="0 1 0"/>
  </joint>
  <joint name="joint4" type="continuous">
    <parent link="link3"/>
    <child link="link4"/>
    <origin rpy="0.0 1.570796325 0.0" xyz="0.0 0.0 0.39225"/>
    <axis xyz="0 1 0"/>
  </joint>
  <joint name="joint5" type="continuous">
    <parent link="link4"/>
    <child link="link5"/>
    <origin rpy="0.0 0.0 0.0" xyz="0.0 0.093 0.0"/>
    <axis xyz="0 0 1"/>
  </joint>
  <joint name="joint6" type="continuous">
    <parent link="link5"/>
    <child link="link6"/>
    <origin rpy="0.0 0.0 0.0" xyz="0.0 0.0 0.09465"/>
    <axis xyz="0 1 0"/>
  </joint>
  <joint name="ee_joint" type="fixed">
    <origin rpy="-1.570796325 0 0" xyz="0 0.0823 0"/>
    <parent link="link6"/>
    <child link="ee_link"/>
  </joint>

<!-- ***** INERTIAL PROPERTIES (LINKS) ***** -->
  <link name="world"/>
  <link name="base_link">
    <inertial>

```

```

    <mass value="4.0"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.0"/>
    <inertia ixx="0.00443333156" ixy="0.0" ixz="0.0"
            iyy="0.00443333156" iyz="0.0" izz="0.0072"/>
  </inertial>
</link>
<link name="link1">
  <inertial>
    <mass value="3.7"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.0"/>
    <inertia ixx="0.010267495893" ixy="0.0" ixz="0.0"
            iyy="0.010267495893" iyz="0.0" izz="0.00666"/>
  </inertial>
</link>
<link name="link2">
  <inertial>
    <mass value="8.393"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.28"/>
    <inertia ixx="0.22689067591" ixy="0.0" ixz="0.0"
            iyy="0.22689067591" iyz="0.0" izz="0.0151074"/>
  </inertial>
</link>
<link name="link3">
  <inertial>
    <mass value="2.275"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.25"/>
    <inertia ixx="0.049443313556" ixy="0.0" ixz="0.0"
            iyy="0.049443313556" iyz="0.0" izz="0.004095"/>
  </inertial>
</link>
<link name="link4">
  <inertial>
    <mass value="1.219"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.0"/>
    <inertia ixx="0.111172755531" ixy="0.0" ixz="0.0"
            iyy="0.111172755531" iyz="0.0" izz="0.21942"/>
  </inertial>
</link>
<link name="link5">
  <inertial>
    <mass value="1.219"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.0"/>
    <inertia ixx="0.111172755531" ixy="0.0" ixz="0.0"
            iyy="0.111172755531" iyz="0.0" izz="0.21942"/>
  </inertial>
</link>
<link name="link6">
  <inertial>
    <mass value="0.1879"/>
    <origin rpy="0 0 0" xyz="0.0 0.0 0.0"/>

```

```

      <inertia ixx="0.0171364731454" ixy="0.0" ixz="0.0"
            iyy="0.0171364731454" iyz="0.0" izz="0.033822"/>
    </inertial>
  </link>
  <link name="ee_link"/>
</robot>

```

Beyond the properties described above, a URDF can describe other properties of a robot, such as its visual appearance (including geometric models of the links) as well as simplified representations of link geometries that can be used for collision detection in motion planning algorithms.

4.3 Summary

- Given an open chain with a fixed reference frame $\{s\}$ and a reference frame $\{b\}$ attached to some point on its last link – this frame is denoted the end-effector frame – the forward kinematics is the mapping $T(\theta)$ from the joint values θ to the position and orientation of $\{b\}$ in $\{s\}$.
- In the Denavit–Hartenberg representation the forward kinematics of an open chain is described in terms of the relative displacements between reference frames attached to each link. If the link frames are sequentially labeled $\{0\}, \dots, \{n+1\}$, where $\{0\}$ is the fixed frame $\{s\}$, $\{i\}$ is a frame attached to link i at joint i (with $i = 1, \dots, n$), and $\{n+1\}$ is the end-effector frame $\{b\}$ then the forward kinematics is expressed as

$$T_{0,n+1}(\theta) = T_{01}(\theta_1) \cdots T_{n-1,n}(\theta_n) T_{n,n+1}$$

where θ_i denotes the joint i variable and $T_{n,n+1}$ indicates the (fixed) configuration of the end-effector frame in $\{n\}$. If the end-effector frame $\{b\}$ is chosen to be coincident with $\{n\}$ then we can dispense with the frame $\{n+1\}$.

- The Denavit–Hartenberg convention requires that reference frames assigned to each link obey a strict convention (see Appendix C). Following this convention, the link frame transformation $T_{i-1,i}$ between link frames $\{i-1\}$ and $\{i\}$ can be parametrized using only four parameters, the Denavit–Hartenberg parameters. Three of these parameters describe the kinematic structure, while the fourth is the joint value. Four numbers is the minimum needed to represent the displacement between two link frames.